

ST. XAVIER'S COLLEGE

MAITIGHAR, KATHMANDU, NEPAL

Post Box :7437

Contact: 4221365,4244636

Email: ktm@sx.edu.np

सेन्ट जेभियर्स कलेज

माईतीघर, काठमाडौं, नेपाल

पो.ब.नं. : ७४३७

फोन : ४२२१३६५, ४२४४६३६

ईमेल : ktm@sx.edu.np

**DEPARTMENT OF COMPUTER SCIENCE****COMPUTER NETWORK****LAB 3****ROUTING IN CISCO PACKET TRACER**

Submitted By:	Submitted To:	Signature:
Name: Siddhant Shrestha CRN: 024bscit040 Semester: 4th Section: B	Er. Sanjay Kumar Yadav Department of Computer Science St. Xavier's College, Maitighar	

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Lab No. 3

Title: Routing in Cisco Packet Tracer

Lab 3.1: Static Routing and DHCP Configuration Between Two Networks

Objective: To configure two separate LANs, each served by its own router-based DHCP pool, and to establish connectivity between the two networks using static routing in Cisco Packet Tracer.

Theory:

When a network includes more than one router, traffic cannot pass between subnets until each router learns how to reach the networks it isn't directly attached to. Static routing offers the simplest way to supply that knowledge — the administrator manually specifies which next-hop address a router should use to reach a given destination network. Unlike dynamic routing protocols, a static route stays fixed and must be updated by hand whenever the topology changes; even so, it remains lightweight, predictable, and well suited to small, stable setups such as the two-router topology used here.

Both routers in this topology also run their own DHCP pool, so the two LANs behind them receive addressing automatically, just as in the earlier lab. In addition, the routers are connected to one another through a separate point-to-point subnet, and it is over this link that the static routes operate so that a host on one LAN can reach a host on the other.

Key Static Routing Concepts

- Destination Network: the remote subnet that the router must learn a path to.
- Subnet Mask: indicates how large the destination network is.
- Next-Hop Address: the interface IP of the neighbouring router that traffic should be forwarded to.
- Static Route Command: `ip route <destination-network> <subnet-mask> <next-hop-address>`

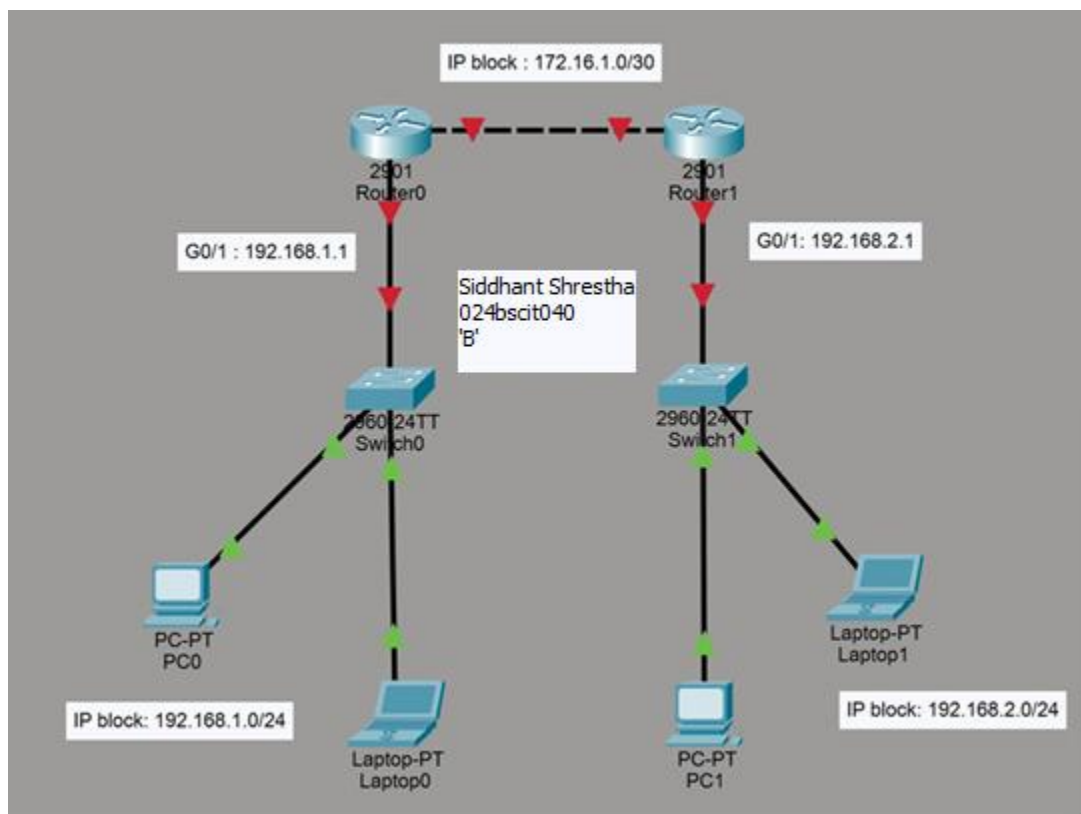
Equipment / Software Needed

- Two units of Router 2901
- Two units of Switch 2960
- PCs and laptops, two devices per LAN
- Straight-through copper cables and a serial/point-to-point cable
- Cisco Packet Tracer software

Procedure:

1. Building the topology

1. Launch Cisco Packet Tracer and add two Router 2901s, two Switch 2960s, and four end devices i.e. two PCs and two laptops.
2. Link Router0 to Switch0, and Switch0 to PC0 and Laptop0, forming LAN 1 (192.168.1.0/24).
3. Link Router1 to Switch1, and Switch1 to PC1 and Laptop1, forming LAN 2 (192.168.2.0/24).
4. Link Router0's GigabitEthernet0/0 port to Router1's GigabitEthernet0/0 port to establish the inter-router connection (172.16.1.0/30).



2. Configuring Router0's Interface

1. Access privileged mode, move into global configuration mode, and assign the hostname:

```
Router>en
```

```
Router#conf t
```

```
Router(config)#hostname Siddhantshresthar0
```

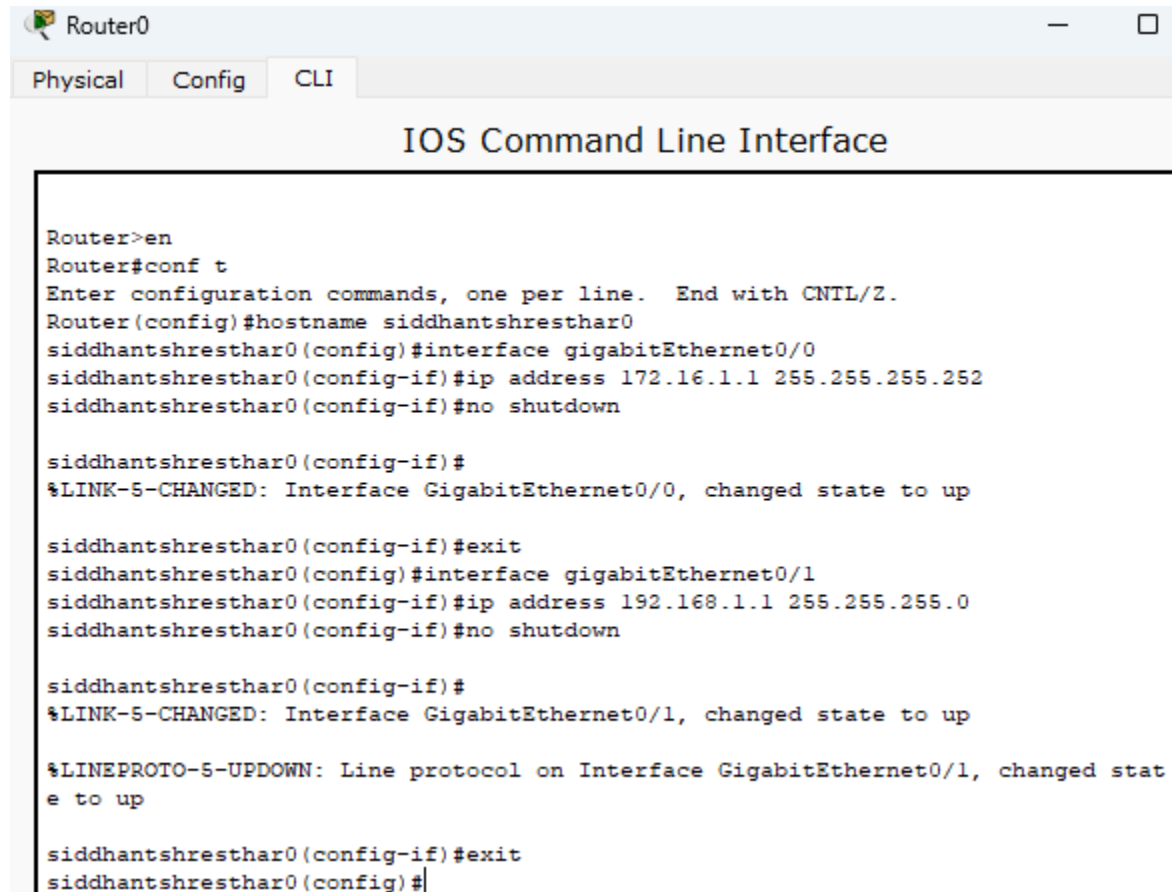
2. Set up the interface facing Router1 over the WAN link:

```
Siddhantshresthar0(config)#interface gigabitEthernet0/0
Siddhantshresthar0(config-if)#ip address 172.16.1.1 255.255.255.252
Siddhantshresthar0(config-if)#no shutdown
```

3. Set up the interface facing the LAN:

```
Siddhantshresthar0(config)#interface gigabitEthernet0/1
Siddhantshresthar0(config-if)#ip address 192.168.1.1 255.255.255.0
Siddhantshresthar0(config-if)#no shutdown
```

OUTPUT:



```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname siddhantshresthar0
siddhantshresthar0(config)#interface gigabitEthernet0/0
siddhantshresthar0(config-if)#ip address 172.16.1.1 255.255.255.252
siddhantshresthar0(config-if)#no shutdown

siddhantshresthar0(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

siddhantshresthar0(config-if)#exit
siddhantshresthar0(config)#interface gigabitEthernet0/1
siddhantshresthar0(config-if)#ip address 192.168.1.1 255.255.255.0
siddhantshresthar0(config-if)#no shutdown

siddhantshresthar0(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

siddhantshresthar0(config-if)#exit
siddhantshresthar0(config)#
```

4. Set up a DHCP pool to serve LAN 1:

```
Siddhantshresthar0(config)#ip dhcp pool LAB_A
Siddhantshresthar0(dhcp-config)#network 192.168.1.0 255.255.255.0
Siddhantshresthar0(dhcp-config)#default-router 192.168.1.1
Siddhantshresthar0(dhcp-config)#dns-server 10.10.10.10
```

5. Store the configuration:

Siddhantshresthar0#wr

OUTPUT:

```
siddhantshresthar0(config)#ip dhcp pool Lab_A
siddhantshresthar0(dhcp-config)#network 192.168.1.0 255.255.255.0
siddhantshresthar0(dhcp-config)#default-router 192.168.1.1
siddhantshresthar0(dhcp-config)#dns-server 10.10.10.10
siddhantshresthar0(dhcp-config)#exit
siddhantshresthar0(config)#exit
siddhantshresthar0#
%SYS-S-CONFIG_I: Configured from console by console

siddhantshresthar0#wr
Building configuration...
[OK]
siddhantshresthar0#
```

3. Configuring Router1

1. Access privileged mode, move into global configuration mode, and assign the hostname:

Router>en

Router#conf t

Router(config)#hostname Siddhantshresthar1

2. Set up the interface that connects to Router0 over the WAN link:

Siddhantshresthar1(config)#interface gigabitEthernet0/0

Siddhantshresthar1(config-if)#ip address 172.16.1.2 255.255.255.252

Siddhantshresthar1(config-if)#no shutdown

3. Set up the LAN-facing interface:

Siddhantshresthar1(config)#interface gigabitEthernet0/1

Siddhantshresthar1(config-if)#ip address 192.168.2.1 255.255.255.0

Siddhantshresthar1(config-if)#no shutdown

OUTPUT:

```
Physical Config CLI
IOS Command Line Interface

^
% Invalid input detected at '^' marker.

Router#
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname siddhantshresthar1
siddhantshresthar1(config)#interface gigabitEthernet 0/0
siddhantshresthar1(config-if)#ip address 172.16.1.2 255.255.255.252
siddhantshresthar1(config-if)#no shutdown

siddhantshresthar1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

siddhantshresthar1(config-if)#exit
siddhantshresthar1(config)#interface gigabitEthernet 0/1
siddhantshresthar1(config-if)#no shutdown

siddhantshresthar1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

siddhantshresthar1(config-if)#exit
```

4. Set up a DHCP pool to serve LAN 2:

```
Siddhantshresthar1(config)#ip dhcp pool LAB_B
```

```
Siddhantshresthar1(dhcp-config)#network 192.168.2.0 255.255.255.0
```

```
Siddhantshresthar1(dhcp-config)#default-router 192.168.2.1
```

```
Siddhantshresthar1(dhcp-config)#dns-server 20.20.20.20
```

5. Store the configuration:

```
Siddhantshresthar1#wr
```

OUTPUT:

```
siddhantshresthar1(config)#ip dhcp pool lab2
siddhantshresthar1(dhcp-config)#network 192.168.2.0 255.255.255.0
siddhantshresthar1(dhcp-config)#default-router 192.168.2.1
siddhantshresthar1(dhcp-config)#dns 20.20.20.20
siddhantshresthar1(dhcp-config)#exit
siddhantshresthar1(config)#exit
siddhantshresthar1#
%SYS-5-CONFIG_I: Configured from console by console

siddhantshresthar1#wr
Building configuration...
[OK]
siddhantshresthar1#
```

4. Verifying that client receive an IP address

1. Open each PC or laptop, go to Desktop → IP Configuration, and select DHCP.
2. Verify that every device picks up an address from its router's DHCP pool.

LAN 1:

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 192.168.1.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 10.10.10.10

Laptop0

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 192.168.1.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 10.10.10.10

LAN 2:

PC1

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 192.168.2.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.2.1

DNS Server 20.20.20.20

Laptop1

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 192.168.2.3

Subnet Mask 255.255.255.0

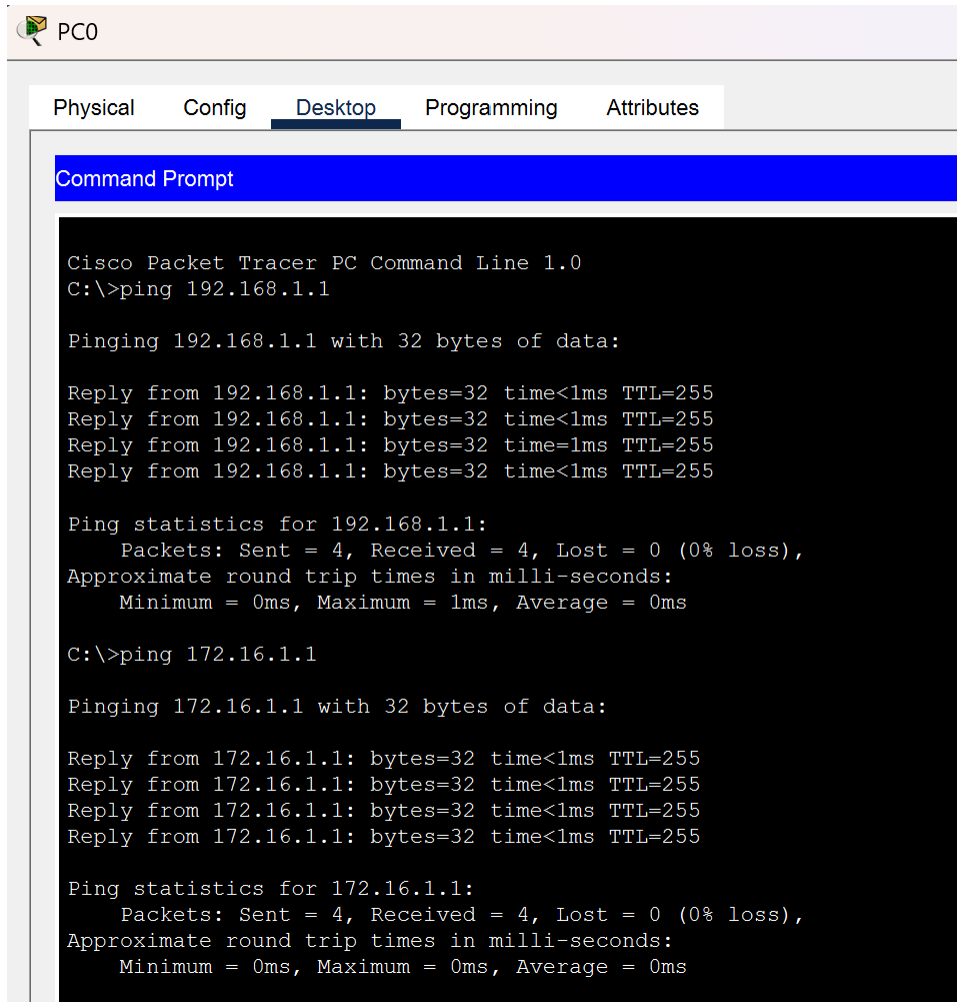
Default Gateway 192.168.2.1

DNS Server 20.20.20.20

5. Checking gateway connectivity

1. From PC0, ping its own default gateway (192.168.1.1) and the directly attached Router0 WAN interface (172.16.1.1) — both should succeed.

OUTPUT:



The screenshot shows the configuration window for PC0 in Cisco Packet Tracer. The 'Desktop' tab is selected, displaying a Command Prompt window. The output shows two successful ping commands from PC0 to 192.168.1.1 and 172.16.1.1, both with 0% loss and 0ms average round trip time.

```
PC0

Physical  Config  Desktop  Programming  Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 172.16.1.1

Pinging 172.16.1.1 with 32 bytes of data:

Reply from 172.16.1.1: bytes=32 time<1ms TTL=255
Reply from 172.16.1.1: bytes=32 time<1ms TTL=255
Reply from 172.16.1.1: bytes=32 time<1ms TTL=255
Reply from 172.16.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.16.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

2. From Router0, ping Router1's WAN interface (172.16.1.2); this attempt fails because neither router has yet learned a route to the other's network.

OUTPUT:

```
C:\>ping 172.16.1.2

Pinging 172.16.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.16.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Note: This result confirms that static routes need to be configured before traffic can pass between the two LANs.

6. Configuring static routing

1. On Router0, check the existing routing table, then add a static route pointing to LAN 2 via Router1's WAN address:

```
Siddhantshresthar0>en
```

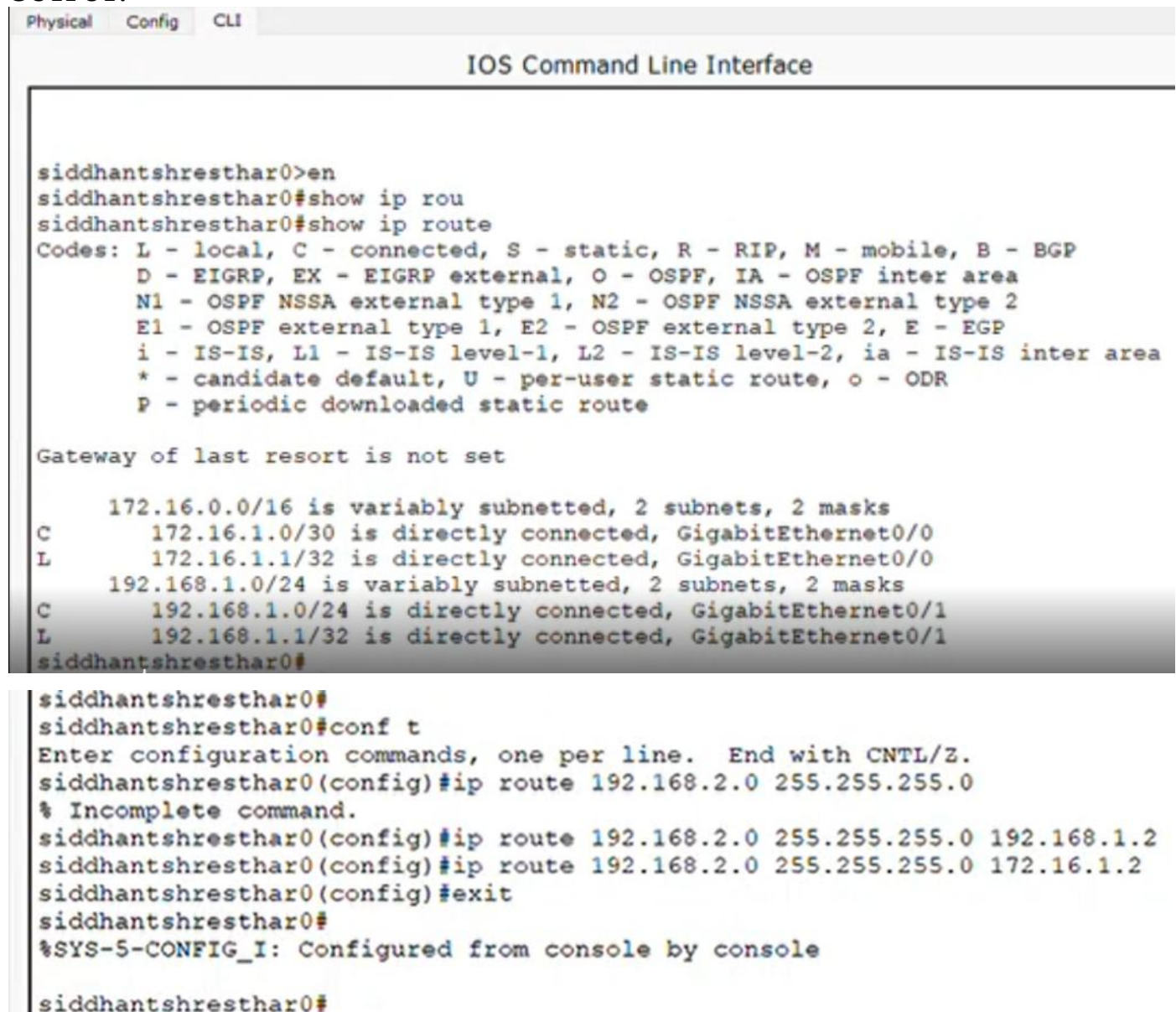
```
Siddhantshresthar0#show ip route
```

```
Siddhantshresthar0#conf t
```

```
Siddhantshresthar0(config)#ip route 192.168.2.0 255.255.255.0 172.16.1.2
```

```
Siddhantshresthar0(config)#exit
```

OUTPUT:



```
Physical  Config  CLI

IOS Command Line Interface

siddhantshresthar0>en
siddhantshresthar0#show ip rou
siddhantshresthar0#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.1.0/30 is directly connected, GigabitEthernet0/0
L       172.16.1.1/32 is directly connected, GigabitEthernet0/0
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/1
L       192.168.1.1/32 is directly connected, GigabitEthernet0/1
siddhantshresthar0#

siddhantshresthar0#
siddhantshresthar0#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
siddhantshresthar0(config)#ip route 192.168.2.0 255.255.255.0
% Incomplete command.
siddhantshresthar0(config)#ip route 192.168.2.0 255.255.255.0 192.168.1.2
siddhantshresthar0(config)#ip route 192.168.2.0 255.255.255.0 172.16.1.2
siddhantshresthar0(config)#exit
siddhantshresthar0#
%SYS-5-CONFIG_I: Configured from console by console

siddhantshresthar0#
```

After the update, the routing table lists: S 192.168.2.0/24 [1/0] via 172.16.1.2

OUTPUT:

```
siddhantshresthar0#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.1.0/30 is directly connected, GigabitEthernet0/0
L       172.16.1.1/32 is directly connected, GigabitEthernet0/0
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/1
L       192.168.1.1/32 is directly connected, GigabitEthernet0/1
S       192.168.2.0/24 [1/0] via 192.168.1.2
                        [1/0] via 172.16.1.2
siddhantshresthar0#
```

2. On Router1, add a static route pointing to LAN 1 via Router0's WAN address:

```
Siddhantshresthar1>e
```

```
n
```

```
Siddhantshresthar1#c
```

```
onft
```

```
Siddhantshresthar1(config)#ip route 192.168.1.0 255.255.255.0 172.16.1.1
```

```
Siddhantshresthar1(config)#exit
```

After the update, the routing table lists: S 192.168.1.0/24 [1/0] via 172.16.1.1

OUTPUT:

```
siddhantshresthar1>en
siddhantshresthar1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
siddhantshresthar1(config)#show ip route

% Invalid input detected at '^' marker.

siddhantshresthar1(config)#ip foute 192.168.1.0 255.255.255.0 172.16.1.1
^
% Invalid input detected at '^' marker.

siddhantshresthar1(config)#ip route 192.168.1.0 255.255.255.0 172.16.1.1
siddhantshresthar1(config)#exit
siddhantshresthar1#
%SYS-5-CONFIG_I: Configured from console by console

siddhantshresthar1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

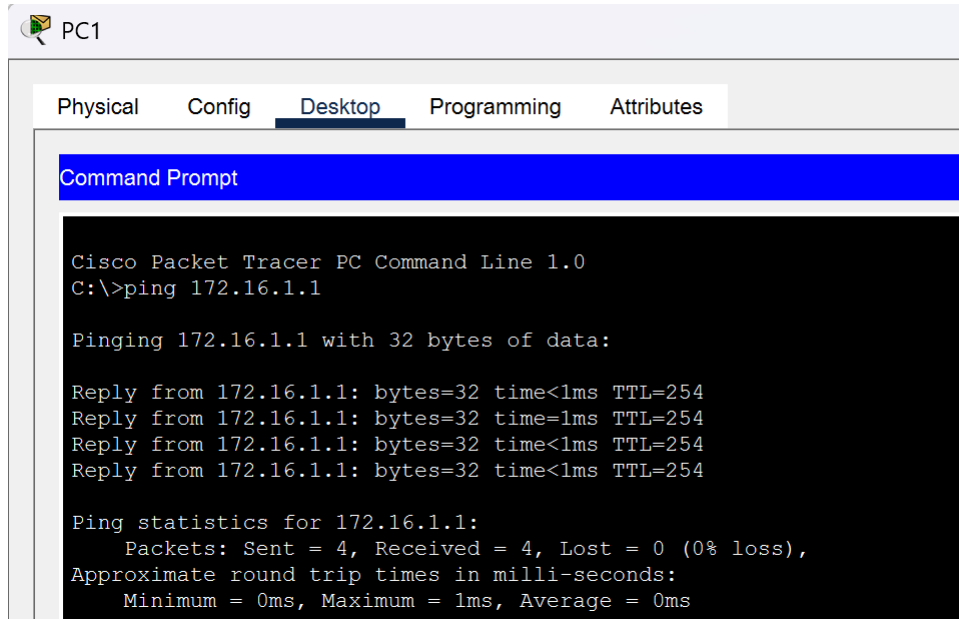
    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.1.0/30 is directly connected, GigabitEthernet0/0
L       172.16.1.2/32 is directly connected, GigabitEthernet0/0
S       192.168.1.0/24 [1/0] via 172.16.1.1
siddhantshresthar1#
```

7. Verifying connectivity after routing

1. From Router1, ping Router0's WAN address and its LAN gateway; both attempts now succeed:

Siddhantshresthar1#ping 172.16.1.1 → Success rate is 100 percent (5/5)

OUTPUT:



```
PC1
Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.1.1

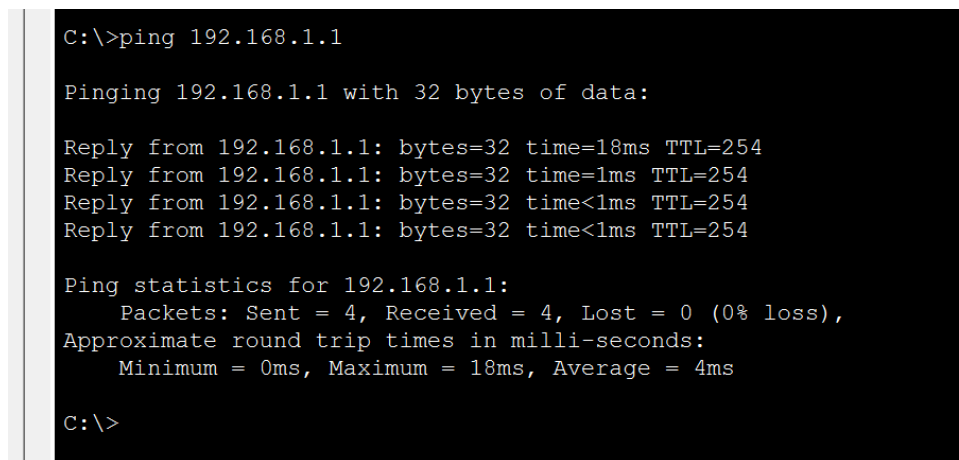
Pinging 172.16.1.1 with 32 bytes of data:

Reply from 172.16.1.1: bytes=32 time<1ms TTL=254
Reply from 172.16.1.1: bytes=32 time=1ms TTL=254
Reply from 172.16.1.1: bytes=32 time<1ms TTL=254
Reply from 172.16.1.1: bytes=32 time<1ms TTL=254

Ping statistics for 172.16.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Siddhantshresthar1#ping 192.168.1.1 → Success rate is 100 percent (5/5)

OUTPUT:



```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=18ms TTL=254
Reply from 192.168.1.1: bytes=32 time=1ms TTL=254
Reply from 192.168.1.1: bytes=32 time<1ms TTL=254
Reply from 192.168.1.1: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 18ms, Average = 4ms

C:\>
```

2. From PC0, ping PC1 (192.168.2.2) to test connectivity across the two LANs:

OUTPUT:

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=2ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=530ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 530ms, Average = 133ms

C:\>|
```

☐ Top

Checking router-to-router reachability directly from the CLI.

OUTPUT:

```
siddhantshresthar1>en
siddhantshresthar1#ping 172.16.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.1.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

siddhantshresthar1#ping 192.168.1.1

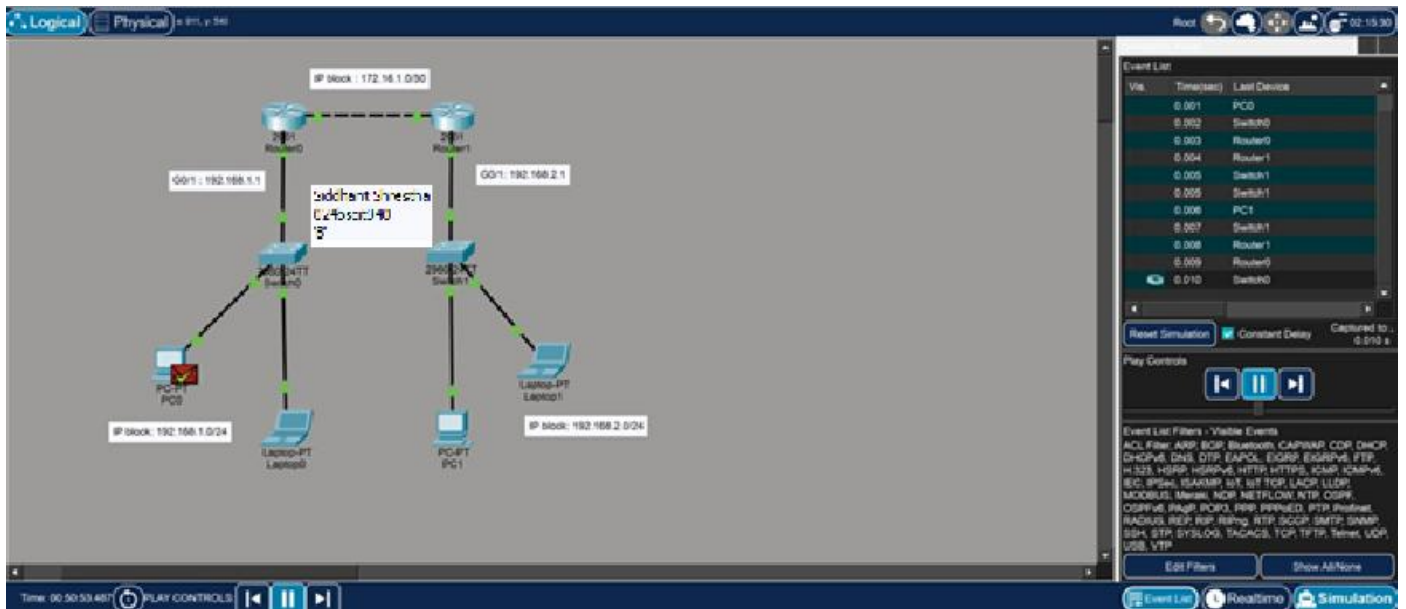
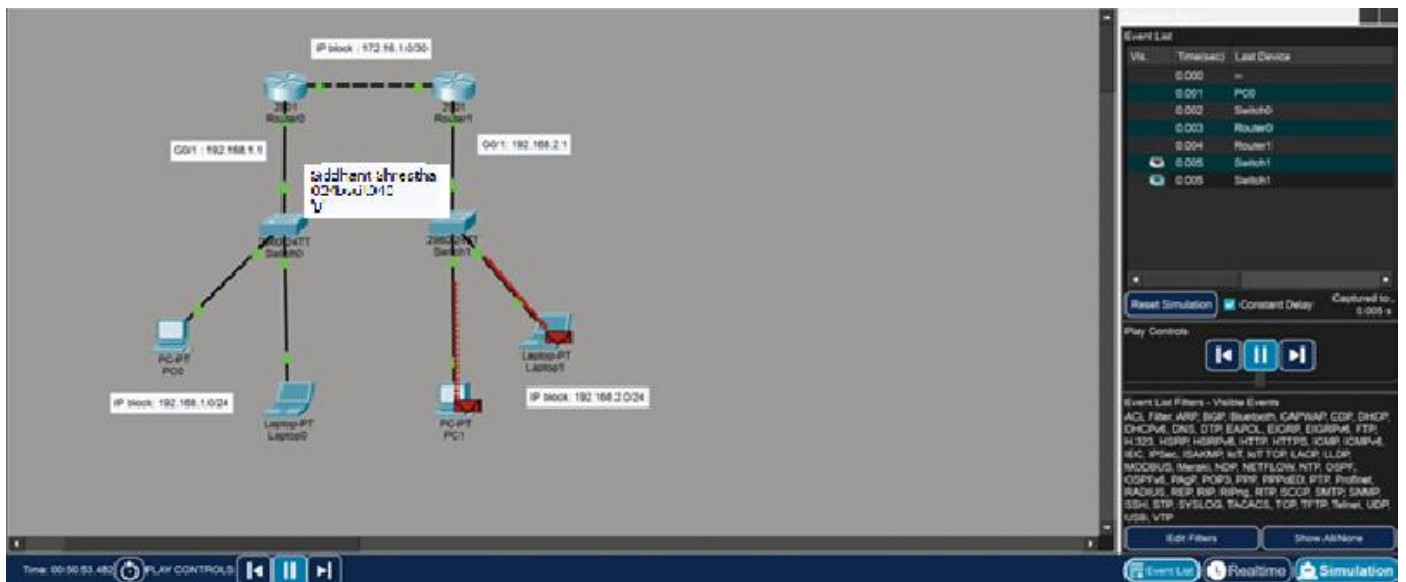
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

siddhantshresthar1#|
```

8. Confirming with Simulation Mode

1. Switch Cisco Packet Tracer into Simulation Mode.
2. Send a ping from PC0 to PC1 and watch the packet move from PC0 → Switch0 → Router0 → Router1 → Switch1 → PC1.
3. Verify that the simulated packets take the statically routed path and that the ping completes successfully.

OUTPUT:



Conclusion:

This lab extended the single-router DHCP setup from the previous lab into a two-router topology, with each router serving its own LAN through an independent DHCP pool. Initial testing confirmed that while each LAN could reach its own gateway, the two routers could not reach one another until static routes were added, illustrating why routing configuration becomes necessary once a network contains more than one router. Once a static route was added on each router pointing to the other's LAN via the shared point-to-point link, connectivity was confirmed end-to-end: first between the two routers, and then between PC0 and PC1 across both LANs, verifying that the static routing setup was working correctly.